

2 The photograph shows kenaf plants that grow in many African countries.



(Source: © Dorling Kindersley Ltd / Alamy Stock Photo)

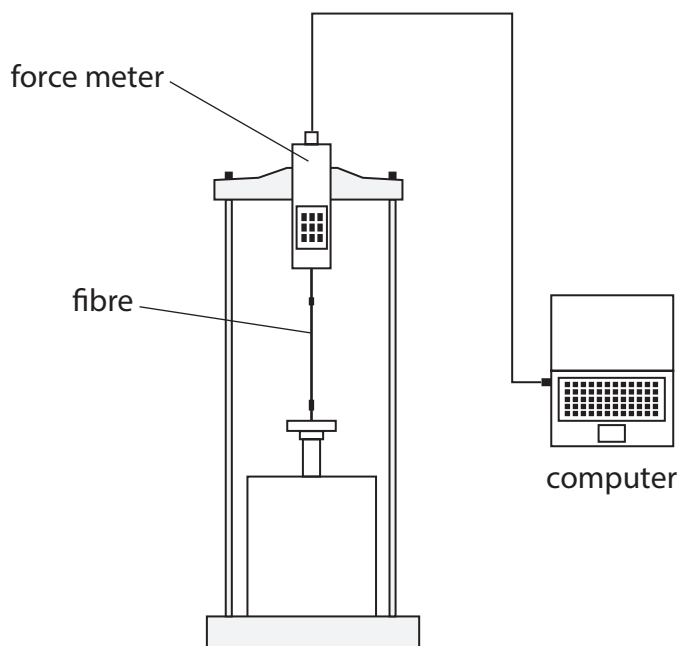
The plants are grown for food and as a source of fibres.

A scientist investigated the tensile strength of samples of fibres from kenaf plants.

Some of the samples of fibres were treated with hydrogen peroxide and other samples were untreated.

The force needed to break each fibre was measured.

The diagram shows the apparatus used.



The apparatus shows a force meter that pulls on the attached fibre. The force needed to break the fibre is recorded on the computer.

(a) Describe how the scientist could reduce **one** risk when using this apparatus.

(2)

Risk

How the scientist could reduce this risk

(b) The table shows the results of this investigation.

Force needed to break a fibre treated with hydrogen peroxide / MPa	Force needed to break an untreated fibre / MPa
894	746
920	821
798	903
964	887
927	795
1021	852
mean 921	mean.....

Calculate the mean percentage increase in tensile strength of fibres treated with hydrogen peroxide compared to untreated fibres.

Give your answer to **three** significant figures.

(3)

.....%

(c) Some variables were controlled in this investigation.

- (i) State **one** abiotic and **one** biotic variable, other than temperature, that could affect this investigation.

(2)

Abiotic variable

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Biotic variable

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- (ii) Choose **one** of the variables you have identified in part (i).

State how this variable can be controlled.

(1)

Variable

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Method of control

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- (d) Sclerenchyma fibres strengthen plant stems.

Describe the structure of a sclerenchyma fibre.

(2)

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(Total for Question 2 = 10 marks)